Maths

1. Integral transforms
    1. Just as a function maps a scalar to a scalar and a functional maps a function to a scalar, an integral transform maps a function to a function. We regard the mapping as an algorithm, but now both input and output are functions. Examples important in image science include the convolution and the Fourier transform.
2. Inverse operator
3. Adjoint operator
    1. adjoint operator, $H^\dagger$ also called conjugate transpose. An operator is Hermitian if $H = H^\dagger$. We say H is self-adjoint if (and on)
4. Hilbert space operators
    1. $g = \mathcal{H}f$
5. Hermitian operators $B^\dagger B$ and $BB^\dagger$.
6. adjoint operator that maps from V to U is the adjoint operator $\mathcal{H}$, denoted $\mathcal{H}^\dagger$. In Hilbert space, we define $\mathcal{H}^\dagger$ such that
    1. $(g_2, \mathcal{H}f_1)_\mathrm{V} = (\mathcal{H}_2^\dagger, f_1)_\mathrm{U}$ for all g_2 and f_1.
7. outer products and projection operators
    1. operator $ba^\dagger$
8. Spectral decomposition of A, shows that the Hermitian matrix A can be expressed as a weighted sum of projection operators, with the weighting coefficients just being the eigenvalues.

References:

1. Harrison H. Barrett, Kyle J. Myers. Foundations of Image Science.